SURYA SANDEEP AKELLA

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EDUCATION

University of Pennsylvania, Philadelphia **Aug 2024 – May 2026**
Master's, Electrical Engineering
Coursework: IoT Edge Computing, Computer Organisation and Design, Hardware Security, Smart Devices, Applied Machine Learning

Gandhi Institute of Technology and Management **Sep 2020 – Apr 2024**
Bachelor's, Electrical Engineering

EXPERIENCE

Electrical Engineer Intern **May 2025 – Aug 2025**
Illinois Tool Works *Troy, OH*

- Designed a custom 2-layer STM32H7-based PCB integrating the ADS131M08 24-bit external ADC, analog front-end circuitry, anti-aliasing filters, and 3.3V/1.8V multi-rail power architecture for high-fidelity audio data acquisition
- Led board bring-up and hardware validation, including power sequencing verification, clock network validation, and debugging of high-speed SPI/I2S interfaces using oscilloscopes and logic analyzers
- Supported investigation of EMI/EMC-related GFCI false-tripping during system validation by examining board-level electrical behavior and possible grounding, filtering, and shielding contributors
- Collaborated with manufacturing on DFM improvements and developed bare-metal C firmware for hardware validation, peripheral control, and embedded audio signal processing support

Firmware QA Engineering Intern **Sep 2025 – Dec 2025**
AMD *Austin, TX*

- Validated hardware platform performance across RDMA, CXL, RoCE, and NVMe interfaces on multi-node Linux systems, characterizing latency, throughput, and electrical behavior across processor generations
- Debugged hardware-firmware integration issues including PCIe link training failures, power delivery anomalies, and memory controller instabilities across BIOS versions using systematic root cause analysis
- Partnered with Platform Engineering teams to identify signal integrity problems and component compatibility issues, contributing to reliable system bring-up and platform validation
- Developed automated hardware test infrastructure and created technical documentation to improve validation coverage, test repeatability, and efficiency across board revisions

PROJECTS

Dynamic Load Balancer for Satellite Microgrid System **Feb 2025 – May 2025**

- Designed 4-layer PCB with dedicated power/ground planes integrating SAMW25 microcontroller, triple-channel high-side current sensing (INA3221), battery fuel gauge (BQ27441), and MOSFET switching network for intelligent load management in satellite communication power systems
- Implemented analog sensing architecture with precision shunt resistors, 3.3V regulation from Li-ion battery, and GPIO-controlled MOSFET drivers; debugged high-side vs. low-side current sensing incompatibility through hardware rework and test point validation
- Performed thermal characterization using infrared imaging, power budget analysis across actuator channels, and developed cloud-connected telemetry system (MQTT/Azure) for real-time current monitoring and battery state-of-charge tracking
- Developed FreeRTOS-based firmware for sensor acquisition and load prioritization while optimizing heap/stack memory allocation to overcome Wi-Fi task constraints in resource-limited embedded environment

Gesture-Controlled Robotic Arm (MimicArm) **Sep 2024 – Nov 2024**

- Designed an embedded system integrating ATmega328PB with MPU6050 MEMS accelerometer, flex sensors, and 4-channel servo driver circuit with ADC input conditioning and I2C interface design
- Implemented modular power distribution system with voltage regulation, decoupling strategy, and ESD protection for robust sensor operation and PWM signal quality optimization
- Developed bare-metal C firmware for sensor data processing and precise gesture-based servo control with low-latency I2C/ADC acquisition
- Validated hardware performance through systematic testing of sensor accuracy, communication bus integrity, and PWM output characteristics

SKILLS

Skills	MATLAB, Python, Altium, Arduino, C/C++, FreeRTOS, PCBA, PCB Design, STM32, Linux, shell
Protocols	SPI, I2C, UART, RTOS, RDMA, CXL, NVMe
Tools	MPLABX, Arduino IDE, Visual Studio Code, Altium, Microchip Studio, STM32CubeIDE, STM32CubeMX, MATLAB, Git, Linux/Unix
Debugging	Oscilloscope, DMM, Function Generator, Logic Analyzer, Spectrum Analyzer, UART Monitor